

# Negation and bias in Czech polar questions: evidence from a naturalness rating study

## Abstract

Negative polar questions (PQs) display bias patterns that have proven difficult to capture within one account. This paper proposes a novel analysis of negation and bias in Czech PQs. While previous literature has distinguished between inner and outer negation readings, we introduce a third, novel category: medial negation. We model the three readings using a unified semantic framework where bias modals operate on the non-at-issue tier, and the negative operator ( $\neg$ ) interacts with them from either the at-issue or non-at-issue level. Inner negation operates at-issue, interacting with a contextual bias modal; medial negation modifies the contextual bias modal on the non-at-issue tier; and outer negation operates non-at-issue and triggers a speaker bias modal. The obligatory focus on the negated modal statement in Czech outer negation PQs triggers exhaustivity, encoding a speaker anti-bias (i.e., that the speaker finds both  $p$  and  $\neg p$  equally accessible) that crucially contrasts with the bias patterns observed in English outer negation PQs. The proposed analysis provides a uniform treatment of negation and question bias while allowing for cross-linguistic variation in question meaning. Finally, we support our theoretical analysis with data from a naturalness judgment task, demonstrating that Czech native speakers are sensitive to all three proposed negation readings.

**Keywords:** negation, bias, polar questions, focus, Czech

# 1 Introduction

Previous research on negation in polar questions (PQs) has shown that it can signal that a PQ is biased, that is, the speaker is not completely ignorant in that she might be biased towards one of the possible answers,  $p$  or  $\neg p$ . Depending on its syntactic position and semantic/pragmatic status, negation has been noticed to cue different flavors of bias, either that which originates from the current context (contextual bias) or from the speaker’s previous epistemic state (speaker bias).

In Czech PQs, negation is represented by the bound verbal prefix *ne-*. Its two possible syntactic positions are shown in (1): (1a) shows a high negation PQ (HiNPQ) with the interrogative word order, while (1b) shows a low negation PQ (LoNPQ) with the declarative word order.

- (1) a. Ne-četl Petr ten e-mail? HiNPQ, interrogative  
 NEG-read Petr DEM e-mail  
 ‘Hasn’t Petr read the e-mail?’  
 b. Petr ne-četl ten e-mail? LoNPQ, declarative  
 Petr NEG-read DEM e-mail  
 ‘Petr has not read the e-mail? / Has Petr not read the e-mail?’

Staňková & Šimík (2025) found that negation in HiNPQs is predominantly interpreted as the so-called outer negation, which gives rise to the inference that the speaker of the PQ thinks  $p$  is likely to hold (i.e., speaker bias for  $p$ ) and which is compatible with the particle *náhodou* (‘by any chance’). Negation in LoNPQs, on the other hand, has been claimed to be ambiguous between the outer vs. inner interpretation, where the latter equals standard sentential negation. LoNPQs have been observed to convey that there is some contextual evidence suggesting  $\neg p$  holds (i.e., contextual bias for  $\neg p$ ).

This paper examines the ambiguity of low negation in PQs like (1b). We show that LoNPQs with an unmarked information structure naturally yield two interpretations: the established inner negation reading, and a second reading that is distinct from outer negation. This second reading – we call it medial negation – cannot be classified as outer negation for two reasons: (i) it strictly conveys a (weak) contextual bias for  $\neg p$ , and (ii) it is incompatible with the particle *náhodou*; see the medial negation PQ (MEDNPQ) in (2). To our knowledge, this paper provides the first identification and systematic empirical evidence of the medial negation reading.

- (2) A: Eva went to the school library yesterday, but I think that she left with her backpack empty.  
 B: Eva si (?? náhodou) ne-půjčila nějakou knížku? MEDNPQ  
 Eva REFL NÁHODOU NEG-borrowed DET.PPI book  
 ‘Eva didn’t borrow some book?’

We show that the outer negation reading becomes available in Czech LoNPQs only under specific structural and prosodic conditions. Once the utterance-initial element is interpreted as a contrastive topic (CT) and the negated verb is focused, the outer reading becomes available – the requirement for contextual bias for  $\neg p$  disappears, and compatibility with *náhodou* is restored; a low outer negation PQ (LoOutNPQ) is shown in (3), where capitals encode sentence stress. Consequently, we argue that low negation ultimately accommodates three distinct readings depending on the information structure of the sentence.

- (3) A: Allegedly, Eva bought a new car.

- B: A [Petr]<sub>CT</sub> si náhodou [NE-KOUPIL]<sub>F</sub> nějaké auto? LoOUTNPQ  
 and Petr REFL NÁHODOU NEG-bought DET.PPI car  
 'And what about Petr, did he buy some car, by any chance?'

Furthermore, we argue that Czech outer negation PQs (OUTNPQs), regardless of whether the negative marker is realized syntactically high or low, exhibits a unique bias profile that distinguishes it from standard descriptions of English outer negation. While English outer negation typically signals (strong) speaker bias for  $p$ , Czech outer negation expresses what we term as speaker anti-bias, that is, the speaker is explicitly considering both  $p$  and  $\neg p$  as accessible. Because of the anti-bias, Czech OUTNPQs sound tentative.

The aim is to develop such an account of negation in Czech PQs that would explain: (i) the bias profiles of negated PQs, and (ii) the licensing conditions each reading of negation imposes on polarity items and certain particles. The analysis builds on previous findings regarding Czech negated PQs but also PQs and negation in general.

The present analysis of negation assumes that the three negation readings – inner, medial and outer – share the same overt realization (*ne-* prefix) and the same semantics ( $\neg$  operator). Where they differ is the meaning tier on which they operate. We draw on the idea of at-issue and non-at-issue meaning tiers (Simons et al. 2010; Beaver et al. 2017; a.o.). In the context of PQs, the ordinary semantic meaning of a PQ is at-issue, while the PQ bias is non-at-issue. We propose that inner negation ( $\neg_{inn}$ ) operates on the at-issue tier, while medial negation ( $\neg_{med}$ ) and outer negation ( $\neg_{out}$ ) operate on the non-at-issue tier. They differ in their scopal relation with respect to two covert modal operators, speaker bias modal ( $\square_{EPI}$ ) and contextual bias modal ( $\square_{EVI}$ ), which are non-at-issue. Inner negation scopes below the contextual bias modal, which delivers the strong contextual bias for  $\neg p$  in inner negation PQs (INNNPQs). Medial negation scopes above the contextual bias modal, which delivers the weak contextual bias for  $\neg p$  in medial negation PQs (MEDNPQs). Outer negation scopes above the speaker bias modal. The focus on the negated modal statement, marked prosodically in LoOUTNPQs or by verb fronting in high outer negation PQs (HiOUTNPQs), triggers the EXH operator, which outputs the speaker anti-bias in OUTNPQs. Their LFs are schematized in (4).

- (4) a.  $[Q [\square_{EVI} [\neg_{inn} p ]]]$  INNNPQ  
 b.  $[Q [\neg_{med} \square_{EVI} [p ]]]$  MEDNPQ  
 c.  $[Q [EXH [\neg_{out} \square_{EPI} [p ]]]$  OUTNPQ

The predictions of the proposal were tested in a naturalness judgment task. The experimental results show that Czech native speakers are able to distinguish between INNNPQs, MEDNPQs and OUTNPQs.

In Section 2 we provide background on some of the basic notions and previous observations made for English: ordinary meaning, bias and negation in PQs. Section 3 presents the empirical background on the form and meaning of Czech PQs. In Section 4 we put forth the semantic analysis for PQ bias and inner, medial and outer negation. Section 5 presents two experiments that were run in order to test some of the predictions of the analysis. In Section 6 we discuss the results of the experiments and how they relate to the formal analysis. Section 7 concludes.

## 2 Background

### 2.1 PQ meaning and bias

Questions are traditionally analyzed as denoting a set of their possible answers (Hamblin 1973; cf. Karttunen 1977 and Groenendijk & Stokhof 1984 for variants thereof). For positive PQs (PPQs), such as *Is John cooking?*, this is the set of the two possible answers,  $\{\lambda w. \text{John is cooking in } w, \lambda w. \text{John is not cooking in } w\}$ , or  $\{p, \neg p\}$  for short, where the first one is called the question radical. For negative PQs, such as *Is John not cooking?*, the question radical would be  $\neg p$  and the set of answers  $\{\neg p, \neg\neg p\}$ . Since  $\neg\neg p$  logically equals  $p$ , the sets of answers for positive and negative PQs are equivalent.

An additional part of PQ meaning is the so-called BIAS (see Romero 2024 for a recent survey). When a PQ is biased, the speaker is not only trying to resolve the  $\{p, \neg p\}$  issue, she is also indicating that one of the possible answers is more likely than the other. Bias can stem from different sources: salient evidence in the current context or the speaker’s prior epistemic state (Büring & Gunlogson 2000; Sudo 2013; a.o.). We will call these notions CONTEXTUAL BIAS and SPEAKER BIAS from now on. The example in (5) illustrates a PQ conveying contextual bias for  $p$  ( $p$  = Carl is left-handed), since there is contextual evidence suggesting  $p$ . Contextual evidence is public and thus mutually available to all participants in the current conversation (Büring & Gunlogson 2000; Goodhue 2024). When considered in isolation, the evidence allows participants to infer  $p$  or increases the likelihood of inferring  $p$ . As Goodhue & Wagner (2018) mention, the evidence can come from various perceptual experiences (visual, auditory).

- (5) *Scenario:* A and B have conducted a psycholinguistic experiment in which the participants have all certified that they are right-handed. They encounter Carl, who they recognize as one of their participants, cutting bread with his left hand.

B: Is Carl left-handed? (Büring & Gunlogson 2000)

The PQ in (6) conveys speaker bias for  $p$ , since speaker B thinks it is possible that Carl is right-handed. Speaker bias is rooted in the epistemic state of the speaker, not directly affected by currently available evidence (Sudo 2013).

- (6) *Scenario:* A and B would like to conduct a psycholinguistic experiment in which all the participants should be right-handed. B previously heard that Carl is right-handed.

B: Is Carl right-handed?

It has been noticed that bias can vary in terms of strength (Šimík forthcoming, Bill & Koev 2023, Bill & Koev 2025). It can be WEAK or STRONG, depending on whether the contextual evidence entails  $p$  (or  $\neg p$ ) or merely suggests  $p$  (or  $\neg p$ ) for contextual bias, and whether the speaker believes  $p$  (or  $\neg p$ ) or merely considers that  $p$  (or  $\neg p$ ) is possible for speaker bias.

In the context of PQs, the set of the possible answers  $\{p, \neg p\}$  is usually claimed to constitute the AT-ISSUE meaning content of a PQ (in the sense of AT-ISSUENESS as discussed by Simons et al. 2010 and Beaver et al. 2017, a.o.). However, previous literature does not give a conclusive answer as to the status of bias. While some proposals include bias in the at-issue import of the PQ (Romero & Han 2004), others more or less explicitly claim that bias is not at-issue (Gutzmann & Miró 2011; Repp 2013; Romero 2015; AnderBois 2019; Frana & Rawlins 2019; Giannakidou & Mari 2025). In the current paper we side with the latter view. In the following section, we discuss the interaction of negation and bias in English PQs and introduce some of the influential approaches to analyzing bias and negation as non-at-issue. We also present a

formal account of Czech (high) outer negation.

## 2.2 Previous research on negation and bias in PQs

Syntactically, negation in English PQs occurs high or low; see (7) which illustrates negation in combination with the two possible word orders of English PQs.<sup>1</sup>

- |     |                         |                      |
|-----|-------------------------|----------------------|
| (7) | a. Isn't John cooking?  | HiNPQ, interrogative |
|     | b. Is John not cooking? | LoNPQ, interrogative |
|     | c. John isn't cooking?  | LoNPQ, declarative   |

According to Ladd (1981) and Romero & Han (2004), there is an asymmetry between English HiNPQs and LoNPQs in that the negative marker in HiNPQs is ambiguous between two readings, inner and outer; while negation in LoNPQs is unambiguously interpreted as inner (cf. e.g., Goodhue (2018) and AnderBois (2019), who claim that English high negation receives only the outer reading). Inner negation represents standard sentential negation which licenses negative polarity items (NPIs), while outer negation represents a notion on another level, which allows for positive polarity items (PPIs) in its scope. Several approaches to outer/inner negation and its relation to bias are introduced in the following paragraphs.<sup>2</sup> We start with the analyses regarding negation and speaker bias, then we turn to negation and contextual bias.

Romero & Han (2004) proposed that HiNPQs host an at-issue conversational epistemic operator, VERUM, where the structural scope between negation and VERUM accounts for both the inner/outer reading dichotomy and polarity item licensing (NPIs/PPIs). According to them, VERUM in English HiOUTNPQs delivers speaker bias for  $p$ . The semantic contribution of outer negation was later re-analyzed by Repp (2013), who introduced a CG-managing counterpart to VERUM called FALSUM. FALSUM operates on the non-at-issue dimension of CG-management content rather than the at-issue answer set, which was later confirmed by Romero (2015). Subsequent authors model speaker bias using distinct non-at-issue semantic operators: Gutzmann & Miró (2011) analyze VERUM as contributing a conventional implicature aimed at resolving the question under discussion; Frana & Rawlins (2019) argue that FALSUM contributes a presupposition; and Giannakidou & Mari (2025) reanalyze outer negation ("Focus-NEG") as a meta-evaluation function over a modal base that introduces a presupposition that  $p$  is preferable to  $\neg p$ . While these non-at-issue approaches successfully prevent bias from interfering with the at-issue meaning, they primarily focus on outer negation and compare it to inner negation only to a limited extent. Furthermore, as we will see in the next section, the Czech empirical landscape of negated PQs is broader, esp. when it comes to LoNPQs. None of these approaches considers medial negation.

Previous literature on contextual bias remains divided. While it is widely accepted that LoNPQs convey a contextual bias for  $\neg p$  (Büring & Gunlogson 2000), the behavior of HiNPQs is heavily debated (Romero 2024), with proposals ranging from compatibility with neutral or  $\neg p$  contexts (Büring & Gunlogson 2000), strict compatibility with only  $\neg p$  evidence (Trinh 2014; Northrup 2014), to compatibility even with evidence for  $p$  (Goodhue 2024). Goodhue (2024) distinguishes between PPQs and LoNPQs through an anaphoric relation: a PQ form

<sup>1</sup>Note on terminology: we consider *interrogatives* and *declaratives* as syntactic objects (i.e., categories of form), while *questions* and *assertions* as speech acts (i.e., categories of meaning) (Huddleston 1994; Gunlogson 2001).

<sup>2</sup>The accounts presented here approach PQ bias through hard-coded semantic operators, but there are other accounts which rely on pragmatic principles when deriving the biases, like Goodhue (2022). The advantages of the semantic approach are: (i) it can capture cross-linguistic variance, (ii) it explains the licensing of polarity items and certain particles in the given PQs. The analysis in the present paper falls into the semantic branch.

must be anaphoric to the proposition made salient in the context. This account predicts that the question radical and its salient antecedent in the discourse are of the same polarity. In Section 3 we will see that this corresponds to the behavior of Czech inner negation PQs, but not for medial negation PQs.

When it comes to the formal analyses of negation and bias in Czech PQs, the landscape is not as rich as that for English. Šimík (forthcoming) puts forth an analysis of Czech high outer negation; e.g., (1a). The main observation here is that Czech high outer negation by default contributes a weak speaker bias for  $p$  (the speaker considers it possible that  $p$  holds). This is based on instances of OUTNPQs where the speaker has no prior strong belief about  $p$ , but where there is an attitude towards  $p$  nevertheless. This is illustrated by the police statement in (8), where the speaker does not believe that the deed has been racially motivated, but rather that it is only a possibility to be considered or explored.

- (8) Ne-víme, jestli (náhodou) ne-šlo o rasově motivovaný čin.  
 NEG-know whether NÁHODOU NEG-went about racially motivated deed  
 ‘We don’t know whether it could have been a racially motivated deed.’  
 (Šimík forthcoming)

The author builds on Repp’s (2013) FALSUM-based analysis of outer negation and postulates a weaker version of it FALSUM<sup>CZ</sup>; its entry is in (9). In contrast to FALSUM, FALSUM<sup>CZ</sup> predicts that the commitment is (i) weaker (involves possibility, not belief), (ii) not necessarily tied to the speaker or the addressee, but possibly also to a linguistically expressed attitude holder (in embedded PQs), (iii) not at-issue, and (iv) not conventionally tied to conversational goals.

$$(9) \quad \llbracket \text{FALSUM}_1^{\text{CZ}} \rrbracket^g(p) = \lambda w : \exists w' \in \text{EPI}_{g(1)}(w)[p(w') = 1]. p \notin \text{CG}_w$$

Šimík (forthcoming) further zooms in on the particle *náhodou* (‘by any chance’). It is observed that it occurs exclusively in OUTNPQs. The author hypothesizes that the particle is a modifier of FALSUM<sup>CZ</sup>’s ordering source. While we believe that Šimík’s (forthcoming) account of outer negation (and of *náhodou*) is on the right track, it is predominantly limited to HINPQs, leaving LoNPQs aside.

To correctly predict the Czech data, the present analysis makes a distinction between what is at-issue and non-at-issue, claiming that both speaker and contextual biases are non-at-issue. On top of the previously discussed inner and outer readings of negation, it argues for a third reading: medial negation. It puts forth that only inner negation is part of the at-issue content, while medial and outer negation operate on the non-at-issue content. Furthermore, it distinguishes Czech outer negation from the English one. It provides a unified analysis of PQ bias using modal semantics as well as a unified analysis of negation, which is formally represented by the logical operator  $\neg$  across the three different readings. The next section presents the empirical landscape of Czech PQs.

### 3 Negation and bias in Czech PQs – empirical background

This section provides empirical background on Czech PQs, their form and meaning. First, word order in PQs in general is discussed. Then we present empirical findings made earlier as well as new data. The section aims to highlight four points about Czech PQs that are crucial for the analysis developed later: (i) word order in Czech can signal the presence of contextual bias or of a contrastive topic, (ii) depending on the syntactic position of the negative marker, negation can receive not only two, but three different readings, (iii) outer negation in Czech

differs from that in English in its distribution and the bias it conveys, and (iv) there are certain particles which can help diagnose the negation reading.

### 3.1 Word order in Czech PQs

Czech PQs can be interrogative (verb-initial) or declarative (verb-non-initial).<sup>3</sup> While interrogative PQs might be used out of the blue and with the intention to resolve true ignorance about  $p$ , declarative PQs in Czech carry a tendency for a contextual bias reading, as suggested by the results of a corpus study by Onoeva & Staňková (2025) (in line with Gunlogson’s 2001 contextual bias condition on English declarative PQs). This is shown in (10) and (11), where the declarative PPQ is felicitous only in a context involving evidence for  $p$ . The examples are adapted from the corpus ORTOFON of spoken Czech (Kopřivová et al. 2024).

- (10) A: Učitel byl přes žně na Křici.  
 teacher was during harvest on Křic  
 ‘The teacher was at Křic during the harvest.’  
 B<sub>1</sub>: Rozvážel učitel vobědy? interrogative PPQ  
 delivered teacher lunches  
 ‘Was the teacher delivering lunch?’  
 B<sub>2</sub>: #Učitel rozvážel vobědy? declarative PPQ  
 teacher delivered lunches  
 ‘The teacher was delivering lunch?’
- (11) A: Učitel byl přes žně na Křici, tak jim rozvážel obědy.  
 teacher was during harvest on Křic so them delivered lunches  
 ‘The teacher was at Křic during the harvest, so he was delivering them lunch.’  
 B<sub>1</sub>: #Rozvážel učitel vobědy? interrogative PPQ  
 delivered teacher lunches  
 ‘Was the teacher delivering lunch?’  
 B<sub>2</sub>: Učitel rozvážel vobědy? declarative PPQ  
 teacher delivered lunches  
 ‘The teacher was delivering lunch?’

Apart from signaling contextual bias, the declarative strategy might be used to encode a contrastive topic (CT). In Czech assertions, CTs have a strong tendency to occur utterance-initially (Sturgeon 2006; Constant 2014). The same tendency is preserved in PQs, where it leads to the declarative word order, as in (12), where apples contrast with plums.<sup>4</sup> PQs with a CT usually appear in contexts without evidence for  $p$  or  $\neg p$  (Romero & Han 2004; Ye 2020; for more Czech examples see Šimík forthcoming). This is also the case for (12), where A provides evidence about  $q$  (where  $q$  = plums are ripe), but crucially not about  $p$  (where  $p$  = apples are ripe).

<sup>3</sup>We are using the term *declarative PQ*, not *rising declarative* – see the overview in Kiss (2024) for the reasons why this is a more suitable term.

<sup>4</sup>In (12), the CT is the subject of the sentence, but object CTs likewise lead to the declarative form, see (i).

- (i) A: Petr sbíral švestky.  
 Petr collected plums  
 ‘Pert was collecting plums.’  
 B: A [jablka]<sub>CT</sub> sbírala Eva?  
 and apples collected Eva  
 ‘And what about the apples, was Eva collecting them?’

(12) *A and B are picking fruit in a garden.*

A: Ty švestky jsou výborné na jídlo právě teď.  
 DEM plums are great for eating right now  
 ‘These plums are ripe now.’

B: A [ty jablka]<sub>CT</sub> už jsou zralá? declarative PPQ  
 and DEM apples already are ripe  
 ‘And what about the apples, are they ripe?’ (Kopřivová et al. 2024)

### 3.2 Negation position and its interpretation

Negation in Czech PQs can be high or low; see (1). Since the negative prefix is a part of the finite verb (as a bound morpheme, it cannot move on its own), the two negation positions map directly onto the two word orders: high negation correlates with the interrogative (verb-initial) order, low negation leads to the declarative. Other combinations are not possible, unlike in English, which can form an interrogative LoNPQ; cf. (7b).

How are HiNPQs and LoNPQs interpreted? Staňková & Šimík (2025) show that there is an asymmetry in this respect. Crucially, this asymmetry goes in a different direction from English: while Czech HiNPQs are unambiguous, LoNPQs allow for ambiguity. To distinguish between the outer vs. inner reading, the authors use testing with PPIs vs. negative concord items (NCIs). Czech is a strict negative concord language where NCIs such as *žádný* ‘no.DET.NCI’ are licensed in syntax (Zeijlstra 2004; Penka 2011), and so present a reliable diagnostic for inner negation. Based on experimental data from Czech native speakers, the authors claim that Czech high negation predominantly receives the outer reading, since HiNPQs with a PPI were rated as significantly more natural than those with an NCI; exemplified in (13a).<sup>5</sup> The authors interpret this result as a clear tendency of negation in HiNPQs to be interpreted as outer.

In LoNPQs, both PPIs and NCIs were rated equally natural; exemplified by (13b). The authors interpret this as low negation being ambiguous between the outer and inner readings. Furthermore, LoNPQs were significantly more natural when following a context implying  $\neg p$  than after neutral context (implying neither  $p$  nor  $\neg p$ ). The authors interpret this result as a clear tendency of LoNPQs to carry contextual bias for  $\neg p$ .

- (13) a. Ne-zasadila tam Dana {nějaké / \*žádné} květiny? HiNPQ  
 NEG-planted there Dana DET.PPI DET.NCI flowers  
 ‘Didn’t Dana plant there some flowers?’  
 b. Dana tam ne-zasadila {nějaké / žádné} květiny? LoNPQ  
 Dana there NEG-planted DET.PPI DET.NCI flowers  
 ‘Dana didn’t plant there some/any flowers?’ (Staňková & Šimík 2025)

### 3.3 Three negation readings

Although we follow Staňková & Šimík (2025) in accepting that LoNPQs can receive the inner negation reading (as evidenced by the licensing of NCIs), we argue that the other reading available in LoNPQs is not an outer negation reading, but that which we term medial negation reading. Both medial negation PQs (MedNPQs) and outer negation PQs (OutNPQs) allow for PPIs in the scope of negation.<sup>6</sup> However, unlike OutNPQs, MedNPQs: (i) are not compatible

<sup>5</sup>In the examples, we gloss positive polarity items as PPI, negative concord items as NCI and negative polarity items as NPI.

<sup>6</sup>PPIs might also occur in inner negation PQs, but must scope over the negative operator; see Section 4.2.



with the particle *náhodou* (‘by any chance’), see (14B); and (ii) strictly require contextual evidence for  $\neg p$  – this condition is met in (14A<sub>1</sub>), so the MEDNPQ is felicitous in such context (marked by ✓), but not in (14A<sub>2</sub>), where the MEDNPQ is infelicitous (marked by #).

(14) ✓ Context with evidence for  $\neg p$ :

A<sub>1</sub>: Eva went to the school library, but I think that she left with her backpack empty.

# Context without evidence for  $p$  or  $\neg p$ :

A<sub>2</sub>: Eva went to the school library and studied there.

B: Eva si (??náhodou) ne-půjčila nějakou knížku? (??náhodou-)MEDNPQ  
 Eva REFL NÁHODOU NEG-borrowed DET.PPI book  
 ‘Eva didn’t borrow some book?’

Both MEDNPQs and inner negation PQs (INNNPQs) obligatorily convey contextual bias for  $\neg p$ ; the bias is strong for the INNNPQs and weak for MEDNPQs. This contrast is shown in (15), where only an INNNPQ is felicitous in a context which entails  $\neg p$ . The INNNPQ is used rhetorically, since it is used in a situation where the interrogative issue has already been settled, which is not possible for MEDNPQs.

(15) A: Petr didn’t buy any cake for the birthday party.

B<sub>1</sub>: Petr ne-koupil žádný dort? INNNPQ  
 Petr NEG-bought DET.NCI cake  
 ‘Petr didn’t buy any cake?’

B<sub>2</sub>: #Petr ne-koupil nějaký dort? MEDNPQ  
 Petr NEG-bought DET.PPI cake  
 ‘Petr didn’t buy some cake?’

The outer negation reading is available in LoNPQs, but only under specific information-structural conditions. Specifically, low outer negation PQs (LoOUTNPQs) are licensed when they contain: (i) focus on the negated finite verb, and (ii) a contrastive topic; see (16).<sup>7</sup> Once these structural requirements are fulfilled, the PQ does not convey contextual evidence implying  $\neg p$ , and it becomes compatible with *náhodou*.

(16) A: Allegedly, Eva bought a new car.

B: A [Petr]<sub>CT</sub> si náhodou [NE-KOUPIL]<sub>F</sub> nějaké auto? LoOUTNPQ  
 and Petr REFL NÁHODOU NEG-bought DET.PPI car  
 ‘And what about Petr, did he buy some car, by any chance?’

Czech OUTNPQs, regardless of whether the negative marker is realized syntactically high or low, exhibit a unique bias profile that distinguishes it from standard descriptions of English OUTNPQs. The latter have been claimed to convey (a relatively strong) speaker bias for  $p$ . However, Czech OUTNPQs have been shown by Šimík (forthcoming) to have a wider distribution and to convey a weaker speaker bias. In Czech OUTNPQs, there is no requirement for the speaker to have a strong prior belief about  $p$ ; the speaker finds both answers likely, which can make OUTNPQs sound more polite or tentative when compared to PPQs. Example (17), adapted from the corpus ORAL v1 (Kopřivová et al. 2017), shows the contrast between a Czech OUTNPQ, a Czech PPQ and an English OUTNPQ. In (17B<sub>1</sub>), the speaker is explicitly considering both possibilities,  $p$  and  $\neg p$ , as accessible. We call this inference ANTI-BIAS on the side of the speaker. It is important that anti-bias does not arise from a clash of speaker bias for  $p$  and contextual bias for  $\neg p$ . The PPQ in (17B<sub>2</sub>) does not convey speaker anti-bias, and, in turn,

<sup>7</sup>In Czech, focus attracts sentence stress (Daneš 1957; Pešková forthcoming).

sounds less tentative in comparison to (17B<sub>1</sub>). English OUTNPQs, like (17B<sub>3</sub>), are infelicitous in such contexts where the speaker does not have any (strong) beliefs about  $p$  or  $\neg p$ .

(17) *A and B are planning a hiking trip in Jizerské hory. B wants to check how long the hike will take and asks A:*

- |                                                                                                                                                                                                                                                                                 |                                                            |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <p>B<sub>1</sub>: Ne-máš tady náhodou mapu?<br/> NEG-have here NÁHODOU map<br/> 'Do you happen to have a map here, by any chance?'</p> <p>B<sub>2</sub>: Máš tady mapu?<br/> have here map<br/> 'Do you have a map here?'</p> <p>B<sub>3</sub>: #Don't you have a map here?</p> | <p>Czech OUTNPQ</p> <p>Czech PPQ</p> <p>English OUTNPQ</p> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|

### 3.4 Compatibility with particles

The three readings of negation can be further distinguished based on their compatibility with three particles: *ještě* ('still'/'yet'), *náhodou* ('by any chance') and *fakt* ('really'). In the following, we introduce the behavior of these particles more generally. Sections 4.2 and 4.3 discuss their compatibility with INNNPQs, MEDNPQs and OUTNPQs and their semantic contribution in more detail.

#### 3.4.1 *Ještě*

In assertions, when co-occurring with atelic predicates, the particle *ještě* ('yet'/'still') can, but does not have to be accompanied by negation; see (18a) and (18b). When co-occurring with telic predicates, *ještě* requires negation; see (18c) and (18d).<sup>8</sup>

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <p>(18) a. Petr ještě spí.<br/> Petr JEŠTĚ sleeps<br/> 'It is still the case that Petr is sleeping.'</p> <p>b. Petr ještě ne-spí.<br/> Petr JEŠTĚ NEG-sleeps<br/> 'It is still the case that Petr isn't sleeping.' / 'Petr isn't sleeping yet.'</p> <p>c. *Petr ještě usnul.<br/> Petr JEŠTĚ fell-asleep<br/> Intended: 'It is still the case that Petr has fallen asleep.'</p> <p>d. Petr ještě ne-usnul.<br/> Petr JEŠTĚ NEG-fell-asleep<br/> 'It is still the case that Petr hasn't fallen asleep.' / 'Petr hasn't fallen asleep yet.'</p> | <p>atelic predicate</p> <p>negative atelic predicate</p> <p>telic predicate</p> <p>negative telic predicate</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|

At least in the structures with telic predicates, we propose to treat *ještě* as an NPI, which requires (inner) negation. As opposed to NCIs, NPIs are assumed to be licensed in semantics/pragmatics (Crnič 2011; Barker 2018) and can be weak or strong, depending on which linguistic environments they occur in: weak NPIs occur in all downward-entailing environments, including the scope of the question operator and the antecedent of conditionals, while strong NPIs occur only in a subset of such environments, including sentential negation (Zwarts

<sup>8</sup> *Ještě* also has an additive meaning (irrelevant for current purposes), which can be translated as 'again' or 'once more'. On that reading, example (18c) is grammatical and can be translated as 'Petr fell asleep again/once more.' For a similar reading of the German *noch* ('still') see Grubic & Wierzbica (2021).

1998). We tentatively propose to treat *ještě* as a strong NPI, based on these facts: (i) it is not licensed by the question operator (19a), (ii) it is not licensed in the antecedent of a conditional (19b), and (iii) it is licensed only by negation (18d). We are not aware of any (formal) analysis of *ještě* in the previous literature.

- (19) a. \*Petr ještě usnul?  
           Petr JEŠTĚ fell-asleep  
           Intended: ‘Is it still the case that Petr has fallen asleep?’  
       b. \*Pokud Petr ještě usnul,       tak musíme být potichu.  
           if       Petr JEŠTĚ fell-asleep then have.to be quiet  
           Intended: ‘If it is still the case that Petr has fallen asleep, then we have to be quiet.’

From now on we will refer to *ještě* in this reading as an NPI and include this in the glosses. In Section 4.2 we will show that because of its NPI-status, *ještě* is predicted to be compatible only with INNMPQs, where negation is at-issue, but not with MEDMPQs and OUTMPQs, where negation is not at-issue.

### 3.4.2 *Náhodou*

That the particle *náhodou* is exclusively compatible with outer negation has been claimed by Staňková & Šimík (2025) and Šimík (forthcoming).<sup>9</sup> We propose that this restricted distribution stems directly from the particle’s semantics. Specifically, *náhodou* facilitates the expression of the speaker anti-bias associated with OUTMPQs. Under our account, the particle widens the quantificational domain of the non-at-issue speaker bias modal within OUTMPQs. By expanding this domain, *náhodou* makes the PQ sound more tentative or polite than its counterpart without *náhodou*. We leave a formalized implementation of this domain-widening analysis for Section 4.3, where it is predicted that *náhodou* is compatible only with OUTMPQs, but not with INNMPQs and MEDMPQs.

### 3.4.3 *Fakt*

When it comes to PPQs, the particle *fakt* (‘really’) is compatible with declarative PPQs conveying contextual bias for *p* (20), but not with interrogative PPQs, where the bias is lacking (21).<sup>10</sup>

- (20) A: Učitel byl přes žně na Křici, tak jim rozvážel obědy.  
           teacher was during harvest on Křic so them delivered lunches  
           ‘The teacher was at Křic during the harvest, so he was delivering them lunch.’  
       B: Učitel fakt rozvážel vobědy? *fakt*-declarative PPQ  
           teacher FAKT delivered lunches  
           ‘Is it really the case that the teacher was delivering lunch?’  
       (21) A: Učitel byl přes žně na Křici.  
               teacher was during harvest on Křic  
               ‘The teacher was at Křic during the harvest.’  
           B: ?Rozvážel fakt učitel vobědy? ?fakt-interrogative PPQ  
               delivered FAKT teacher lunches  
               Intended: ‘Is it really the case that the teaches was delivering lunch?’

<sup>9</sup>In assertions, *náhodou* can be used as an adverb meaning ‘accidentally’.

<sup>10</sup>*Fakt* has an additional meaning (irrelevant for current purposes), upon which it modifies a lexically gradable predicate, similarly to the English *really* (Partee 2004; Bill & Koev 2023).

These data show that *fakt* patterns with the presence of contextual bias in PQs. While *fakt* has not, to our knowledge, received any (formal) analysis in previous literature, we propose that *fakt* targets the non-at-issue layer of contextually biased PQs and loosens the ordering source of the contextual bias modal. The quantificational domain of the modal is widened so that it includes less stereotypical worlds. As a result, PQs with *fakt* like (20) invite the addressee to consider even less stereotypical worlds with evidence for  $p$  before answering. We will come back to *fakt* and its interaction with the contextual bias modal in Section 4.3, where it is predicted that *fakt* is compatible with INNPNQs and MEDPNQs, but not OUTPNQs.

The upshot of the previous findings and claims about Czech negative PQs is that high negation tends to be interpreted as outer, while low negation is ambiguous. The novel data presented above showed that we can distinguish between three readings of negation in Czech: inner, medial and outer. The present article aims to develop a novel analysis of PQ bias, which captures its type (contextual vs. speaker), polarity (for  $\neg p$  or for  $p$ ) and strength (weak vs. strong) in connection to negation in Czech PQs. This analysis takes all three observed readings of negation and makes predictions as to which inferences they contribute and what material they license. In the upcoming section, we propose that inner negation operates on the at-issue level (just like standard sentential negation in assertions), while medial and outer negation operate on the non-at-issue level. Inner and medial negation interact with the (non-at-issue) contextual bias modal, while outer negation triggers and interacts with the (non-at-issue) speaker bias modal. Semantically, all three negations are represented by the same logical operator  $\neg$ , while both types of bias are represented by covert modal operators. Crucially, only one of the negative operators can be active in a given sentence (i.e., they are in complementary distribution). As for the bias profiles of Czech negated PQs, the interaction of the negative operator with the respective bias modal delivers the empirically attested inferences.

## 4 Proposed analyses

In this section, we flesh out the proposed analysis of PQ bias and PQ negation in Czech. First, we introduce two covert modal operators contributing speaker and contextual bias. Each bias type is modeled by a distinct modal operator (in the sense of Kratzer 1991). The denotation of the speaker bias modal is in (22) and the denotation of the contextual bias modal is in (23). In what follows we write the modals as  $\Box_{\text{EPI}}$  and  $\Box_{\text{EVI}}$ , respectively. Both contribute to the non-at-issue dimension, while leaving the at-issue content intact.<sup>11</sup>

$$(22) \quad \llbracket \Box_{\text{EPI}} \rrbracket^{g,o} = \text{speaker bias modal}$$

At-issue:  $\lambda q . \lambda w . q(w)$   
Non-at-issue:  $\lambda q . \lambda w . \forall w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [q(w') = 1]$   
where  $g(1)$  is the attitude holder;  $\text{EPI}_{g(1)}(w)$  is the set of worlds compatible with what  $g(1)$  knows and believes in  $w$ ;  $\max_{o(w)}(\text{EPI}_{g(1)}(w))$  selects the most ideal worlds from  $\text{EPI}_{g(1)}(w)$ , given the ordering by the stereotypical ordering source  $o(w)$ ; and  $q$  is the question radical

Informally:

Based on what the speaker believes, and assuming things are following their stereotypical course, it must be that  $p$  holds.

<sup>11</sup>In the current analysis, we remain silent on the issue whether PQ bias should be classified as a presupposition or a conventional implicature. Treating it simply as non-at-issue suffices to give the right predictions, but see for example Korotkova (2023) for a discussion on this.

- (23)  $\llbracket \Box_{\text{EVI } 1} \rrbracket^{g,o} =$  contextual bias modal  
 At-issue:  $\lambda q . \lambda w . q(w)$   
 Non-at-issue:  $\lambda q . \lambda w . \forall w' \in \max_{o(w)}(\text{EVI}_{g(1)}(w)) [q(w') = 1] \}$   
 where  $g(1)$  is the attitude holder;  $\text{EVI}_{g(1)}(w)$  is the set of worlds compatible with what  $g(1)$  has evidence for in  $w$ ;  $\max_{o(w)}(\text{EVI}_{g(1)}(w))$  selects the most ideal worlds from  $\text{EVI}_{g(1)}(w)$ , given the ordering by the stereotypical ordering source  $o(w)$ ; and  $q$  is the question radical

Informally:

Based on the current contextual evidence, and assuming things are following their stereotypical course, it must be that  $p$  holds.

The analysis assumes that the speaker bias modal obligatorily arises once outer negation is active, and that it occurs in the scope of the exhaustivity operator (more on this in Section 4.1.3). On the other hand, the contextual bias modal can occur independently of negation (in PPQs conveying contextual bias for  $p$ ). In the following subsection, we show how negation interacts with these two modals.

## 4.1 Obligatory biases

In Section 3.3 we observed that INNNPQs convey strong contextual bias for  $\neg p$ , MEDNPQs convey weak contextual bias for  $\neg p$  and OUTNPQs convey speaker anti-bias. In the following we show how the biases are derived.

### 4.1.1 Inner negation polar questions

We put forth that in INNNPQs, the negation operator  $\neg_{\text{inn}}$  is part of the question radical and modifies the at-issue content only. The contextual bias modal scopes above the question radical. The meaning of an INNNPQ is in (24).<sup>12</sup>

- (24)  $\llbracket [Q [\Box_{\text{EVI}} [\neg_{\text{inn}} p]]] \rrbracket =$   
 At-issue:  $\{\lambda w . \neg p(w), \lambda w . \neg \neg p(w)\} \equiv \{\lambda w . p(w), \lambda w . \neg p(w)\}$   
 Non-at-issue:  $\lambda w . [\forall w' \in \max_{o(w)}(\text{EVI}_{g(1)}(w)) [\neg p(w') = 1]]$

The non-at-issue content in (24) could be informally formulated as ‘based on the current contextual evidence, and assuming things are following their stereotypical course, it must be that  $\neg p$  holds’, where strong necessity is encoded by the universal quantifier. This delivers the strong contextual bias for  $\neg p$  in INNNPQs.

### 4.1.2 Medial negation polar questions

MEDNPQs are proposed to contain the contextual bias modal on the non-at-issue level, just like INNNPQs. However, the negation operator  $\neg_{\text{med}}$  scopes over the contextual bias modal and modifies it.  $\neg_{\text{med}}$  operates directly on the non-at-issue content, leaving what is at-issue intact (the question radical remains  $p$ ). The meaning of a MEDNPQ is in (25)

- (25)  $\llbracket [Q [\neg_{\text{med}} \Box_{\text{EVI}} [p]]] \rrbracket =$

<sup>12</sup>Similarly to the approach of Romero (2015), we take the Q morpheme to introduce both the at-issue as well as the non-at-issue content of a PQ. We also assume that one non-at-issue content can modify another non-at-issue content (see also Gutzmann 2017).

At-issue:  $\{\lambda w. p(w), \lambda w. \neg p(w)\}$   
Non-at-issue:  $\lambda w. [\neg \forall w' \in \max_{o(w)}(\text{EVI}_{g(1)}(w))[p(w') = 1]]$

The non-at-issue content in (25) could be informally formulated as ‘based on the current contextual evidence and assuming things are following their stereotypical course, it must not be that  $p$  holds’. How is the weak bias semantics in MEDNPQs delivered? By DeMorgan’s law, negation scoping over a universal quantifier is equivalent to an existential quantifier scoping over negation. At the non-at-issue level,  $\neg_{med}$  scopes over the universal quantifier ( $\neg_{med} > \forall$ ); by the equivalence in (26) we arrive at the interpretation involving an existential quantifier scoping over the negation ( $\exists > \neg_{med}$ ), leading to the inference of a weak contextual bias. Informally, this inference says that ‘based on the current contextual evidence, it is possible that  $\neg p$  holds’.

$$(26) \quad \lambda w. [\neg \forall w' \in \max_{o(w)}(\text{EVI}_{g(1)}(w))[p(w') = 1]] \\ \equiv \lambda w. [\exists w' \in \max_{o(w)}(\text{EVI}_{g(1)}(w))[\neg p(w') = 1]]$$

Semantically, the weak contextual bias for  $\neg p$  in MEDNPQs is consistent with strong contextual bias for  $\neg p$ . This would suggest that contextual bias of MEDNPQs might be ambiguous between being weak and strong. However, since INNNPQs necessarily carry strong contextual bias for  $\neg p$ , this inference is blocked for MEDNPQs. In this sense, MEDNPQs and INNNPQs are in complementary distribution.

### 4.1.3 Outer negation polar questions

We put forth that in OUTNPQs, the negation operator  $\neg_{out}$  triggers the presence of the speaker bias modal on the non-at-issue level and scopes above it. Similarly to  $\neg_{med}$ ,  $\neg_{out}$  modifies the non-at-issue content, leaving the question radical  $p$  intact. The modal statement is focused. An exhaustification operator EXH is triggered above the focused modal, as defined in (27) (based on Fox 2007). EXH applies to a proposition  $a$  and a set of formal alternatives  $Alt(a)$  and its effect is equivalent to asserting  $a$  while negating all of  $a$ ’s innocently excludable (IE) alternatives, as defined in (28).

$$(27) \quad \llbracket \text{EXH} \rrbracket(Alt(a))(a)(w) \text{ iff } a(w) \wedge \forall a' \in Alt_{IE}(a, Alt(a))[\neg a'(w)]$$

$$(28) \quad Alt_{IE}(a, Alt(a)) = \bigcap \{Alt(a)' \subseteq Alt(a) : Alt(a)' \text{ is a maximal set in } Alt(a), \text{ s.t. } \{\neg a' : a' \in Alt(p)'\} \cup \{a\} \text{ is consistent}\}$$

The meaning of an OUTNPQ is captured in (29) (note that this applies to HiOUTNPQs as well as to LoOUTNPQs). The modal statement  $\phi$  is focused and projects a set of formal alternatives  $Alt(\phi)$ . Importantly, it is the modal statement, not polarity, which is focused; overtly this is cued by focus on the negated finite verb. The focusing is not optional, but obligatory, as  $\phi$  and  $Alt(\phi)$  are required by EXH. EXH applies to  $\phi$  and  $Alt(\phi)$  on the non-at-issue level.<sup>13</sup> The strengthened meaning EXH outputs is equivalent to the inference which could be informally paraphrased as ‘based on what the speaker believes, and assuming things are following the stereotypical course, it is possible that  $p$  as well as  $\neg p$  holds’. This is the anti-bias conveyed by OUTNPQs.

$$(29) \quad \llbracket [Q [\text{EXH} [\neg_{out} \Box_{EPI} p]]] \rrbracket =$$

<sup>13</sup>In our analysis, we assume that EXH attaches locally and operates at the non-at-issue level. Although this assumption is rather non-standard, there have been proposals pointing in the same direction, see Magri (2009); Gajewski & Sharvit (2011); Marty (2017); Marty & Romoli (2021).

At-issue:  $\{\lambda w . p(w), \lambda w . \neg p(w)\}$

Non-at-issue ordinary meaning:

$$\begin{aligned} & \lambda w . [\neg \forall w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [p(w') = 1]] \\ & \equiv \lambda w . [\exists w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]] \end{aligned} \quad (= \phi)$$

Non-at-issue focus meaning:

$$\begin{aligned} & \{\lambda w . [\exists w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]], \\ & \lambda w . [\forall w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]]\} \end{aligned} \quad (= \text{Alt}(\phi))$$

Non-at-issue EXH output:

$$\begin{aligned} & \text{EXH}(\text{Alt}(\phi))(\phi) = \\ & \lambda w . [\exists w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]] \wedge \\ & \lambda w . [\neg \forall w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]] \\ & \equiv \lambda w . [\exists w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [\neg p(w') = 1]] \wedge \\ & \lambda w . [\exists w' \in \max_{o(w)}(\text{EPI}_{g(1)}(w)) [p(w') = 1]] \end{aligned}$$

#### 4.1.4 Optional biases

A short note on the biases Czech NPQs can, but do not have to convey. Our analysis predicts that the interpretation of INNNPQs and MEDNPQs does not involve speaker bias, that is, they might convey speaker bias for  $p$ , for  $\neg p$ , or for neither  $p$  or  $\neg p$ ; see (30).

- (30) ✓ Prior speaker belief for  $p$ :  
*B knows that Petr and Eva are good friends.*
- ✓ Prior speaker belief for  $\neg p$ :  
*B knows that Petr doesn't have much money right now.*
- ✓ Prior speaker ignorance about  $p$  and  $\neg p$ :  
*B doesn't know whether Petr and Eva are friends. B hasn't seen Petr for years.*
- A: Petr didn't buy any birthday gift for Eva.
- B<sub>1</sub>: Petr ne-koupil Evě žádný dárek? INNNPQ  
Petr NEG-bought Eva DET.NCI cake  
'Petr didn't buy any birthday gift for Eva?'
- B<sub>2</sub>: Petr ne-koupil Evě nějaký dárek? MEDNPQ  
Petr NEG-bought Eva DET.PPI cake  
'Petr didn't buy some birthday gift for Eva?'

Furthermore, the interpretation of OUTNPQs does not involve contextual bias. The data in (31) show that OUTNPQs with high negation (HiOUTNPQs) might convey contextual bias for  $p$ , for  $\neg p$ , or for neither  $p$  or  $\neg p$ , respectively.

- (31) ✓ Contextual evidence for  $p$ :  
A<sub>1</sub>: Allegedly, Petr bought a gift for Eva.
- ✓ Contextual evidence for  $\neg p$ :  
A<sub>2</sub>: Allegedly, Petr bought a bouquet for Eva.
- ✓ Context without evidence for  $p$  or  $\neg p$ :  
A<sub>3</sub>: Allegedly, Petr enjoyed Eva's birthday party.
- B: Ne-koupil Petr Evě nějakou knížku? HiOUTNPQ  
NEG-bought Petr Eva DET.PPI book  
'Hasn't Petr bought some book for Eva?'

However, (33) shows that OUTNPQs with low negation (LoOUTNPQs) cannot convey contextual bias for  $p$  or  $\neg p$ . This is because LoOUTNPQs involve contrastive topics, which typically occur in contexts without evidence for  $p$  or  $\neg p$  and require a contrasting element in the preceding discourse. Thus, HiOUTNPQs are predicted to have wider distribution than LoOUTNPQs.

(32) # Contextual evidence for  $p$ :

A<sub>1</sub>: Allegedly, Petr bought a gift for Eva.

# Contextual evidence for  $\neg p$ :

A<sub>2</sub>: Allegedly, Petr bought a bouquet for Eva.

# Context without evidence for  $p$  or  $\neg p$ :

A<sub>3</sub>: Allegedly, Petr enjoyed Eva's birthday party.

✓ Context without evidence for  $p$  or  $\neg p$  and with a contrasting element:

A<sub>4</sub>: Allegedly, Adam bought a book for Eva.

B: A [Petr]<sub>CT</sub> Evě NEKOUPIL nějakou knížku? LoOUTNPQ  
 and Petr Eva NEG-bought DET.PPI book  
 'And what about Petr, hasn't he bought some book for Eva?'

## 4.2 Polarity items

In Sections 3.2 and 3.3 we observed that inner negation licenses NCIs, while medial and outer negation allow for PPIs in their scope. In our account, we assume that only inner negation is part of the question radical and behaves like standard sentential negation, which includes its syntactic as well as semantic behavior. Following the broadly accepted account of negation syntax by Zeijlstra (2004), we assume that the negative marker *ne-* agrees with the semantic negation operator which sits in NegP above  $\nu P$ . It is the operator which carries the semantic negation, not the marker itself. The negated verb carries the [uNeg] feature which checks with the [iNeg] feature on the operator. NCIs carry [uNeg], too, and so are licensed under  $\neg_{inn}$ ; see (33a). Further,  $\neg_{inn}$  is predicted to license NPIs in its scope. In case the predicate of the INNNPQ is telic,  $\neg_{inn}$  is predicted to license the NPI particle *ještě* ('still'/'yet'); see (33a). PPIs are predicted to scope only over  $\neg_{inn}$ , but not under it; see (33b).

- (33) a. Petr ještě ne-složil žádnou zkoušku?  $\neg_{inn} > \exists_{NCI}$   
 Petr JEŠTĚ.NPI NEG-passed DET.NCI exam  
 'Is it still the case that Petr hasn't passed any exam?' / 'Petr hasn't passed any exam yet?'
- b. Petr ještě ne-složil nějakou zkoušku?  $\exists_{PPI} > \neg_{inn} / * \neg_{inn} > \exists_{PPI}$   
 Petr JEŠTĚ.NPI NEG-passed DET.PPI exam  
 'Is it still the case that there is an exam that Petr hasn't passed?' / 'There is an exam that Petr hasn't passed yet?'
- #'Is it still the case that Petr hasn't passed some exam?'

That inner negation licenses NPIs is in line with previous accounts: Romero & Han (2004) diagnose inner negation by the NPI *either* in English NPQs, while Repp & Geist (2025) use the NPI *eščě* 'yet' in Russian (but see AnderBois 2019 for arguments against using NPIs as diagnostics of inner negation).

Since  $\neg_{med}$  and  $\neg_{out}$  operate on the non-at-issue level, they are predicted to allow for PPIs in their scope; see (34a) and (34b). For the same reason,  $\neg_{med}$  and  $\neg_{out}$  are predicted to anti-license NCIs and NPIs, including *ještě* in the case of telic predicates; see (34c-f).



- (34) *Eva and her colleagues from the office ordered take-out food for lunch. Eva is a vegetarian. The delivery guy brings only pizzas with ham. One of the colleagues asks:*
- a. Eva si ne-objednala nějaký salát?  $\neg_{med} > \exists_{PPI}$   
 Eva REFL NEG-ordered DET.PPI salad  
 ‘Eva didn’t order some salad?’
  - b. Ne-objednala si Eva nějaký salát?  $\neg_{out} > \exists_{PPI}$   
 NEG-ordered REFL Eva DET.PPI salad  
 ‘Didn’t Eva order some salad?’
  - c. Eva si ne-objednala někde nic jiného?  $*\neg_{med} > \exists_{NCI}$   
 Eva REFL NEG-ordered somewhere.PPI nothing.NCI else  
 (no translation)
  - d. Ne-objednala si Eva někde nic jiného?  $*\neg_{out} > \exists_{NCI}$   
 NEG-ordered REFL Eva somewhere.PPI nothing.NCI else  
 (no translation)
  - e. ??Eva si ještě ne-objednala nějaké jídlo?  $??ještě-MEDNPQ$   
 Eva REFL JEŠTĚ.NPI NEG-ordered DET.PPI food  
 (no translation)
  - f. ??Ne-objednala si ještě Eva nějaké jídlo?  $??ještě-OUTNPQ$   
 NEG-ordered REFL JEŠTĚ.NPI Eva DET.PPI food  
 (no translation)

### 4.3 Particles

In Section 3.4 we observed that *náhodou* (‘by any chance’) is compatible only with OUTNPQs, while *fakt* (‘really’) is compatible with INNNPQs and MEDNPQs (and PPQs), but not with OUTNPQs. In the following we put forth our formal analysis of these two particles.

#### 4.3.1 Felicity of *náhodou*

We put forth that *náhodou* loosens the ordering source of  $\Box_{EPI}$ . According to our analysis,  $\Box_{EPI}$  is an integral part of OUTNPQ meaning, but is missing in MEDNPQs and INNNPQs. As *náhodou* requires  $\Box_{EPI}$  to operate on, it is predicted to be incompatible with MEDNPQs, as already shown in (14), and with INNNPQs, as shown in (35).

- (35) A: Eva went to the school library, but I think that she left with her backpack empty.  
 B: Eva si (??náhodou) ne-půjčila žádnou knížku? (??náhodou-)INNNPQ  
 Eva REFL NÁHODOU NEG-borrowed DET.NCI book  
 ‘Eva didn’t borrow any book?’

Following the original proposal for *náhodou* by Šimík (forthcoming), we assume its function is to widen the quantificational domain, in our case it is the domain of  $\Box_{EPI}$ . *Náhodou* loosens the ordering source, which leads to the inclusion of stereotypical as well as less stereotypical worlds in the domain of the modal. This way *náhodou* makes the inference of anti-bias even more evident than it is in OUTNPQs without *náhodou*. For example, in (17), repeated here as (36), the speaker considers  $p$  and  $\neg p$  as accessible answers in the stereotypical as well as less stereotypical worlds.

- (36) *A and B are planning a hiking trip in Jizerské hory. B wants to check how long the hike will take and asks A:*

- B: Ne-máš tady náhodou mapu? náhodou-HiOUTNPQ  
 NEG-have here NÁHODOU map  
 'Do you happen to have a map here, by any chance?'

The meaning of (17)/(36) is captured in (37), where  $o'(w)$  is the loosened stereotypical ordering source:

$$(37) \quad \llbracket (17)/(36) \rrbracket =$$

At-issue:  $\{\lambda w. \text{HAVE}(A, \text{MAP})(w), \lambda w. \neg \text{HAVE}(A, \text{MAP})(w)\}$   
 Non-at-issue EXH output:  
 $\lambda w. [\exists w' \in \max_{o'(w)}(\text{EPI}_B(w)) [\neg \text{HAVE}(A, \text{MAP})(w') = 1]] \wedge$   
 $\lambda w. [\exists w' \in \max_{o'(w)}(\text{EPI}_B(w)) [\text{HAVE}(A, \text{MAP})(w') = 1]]$

#### 4.3.2 Felicity of *fakt*

In Section 3.4, we suggested that *fakt* loosens the ordering source of  $\Box_{\text{EVI}}$ . According to our analysis above,  $\Box_{\text{EVI}}$  is grammatically encoded in INNNPQs and MEDNPQs, but not in OUTNPQs. *Fakt* requires  $\Box_{\text{EVI}}$  to operate on to be used felicitously, and so *fakt*-INNNPQs (38a) and *fakt*-MEDNPQs (38b) are predicted to be felicitous, while *fakt*-OUTNPQs (both *fakt*-HiOUTNPQs (38c) and *fakt*-LoOUTNPQs (38d)) are predicted to be infelicitous.

- (38) a. Eva si fakt ne-půjčila žádnou knížku? fakt-INNPQ  
 Eva REFL FAKT NEG-borrowed DET.NCI book  
 'Eva really hasn't borrowed any book?'  
 b. Eva si fakt ne-půjčila nějakou knížku? fakt-MEDNPQ  
 Eva REFL FAKT NEG-borrowed DET.PPI book  
 'Is it really the case that Eva hasn't borrowed some?'  
 c. ??Ne-půjčila si Eva fakt nějakou knížku? ??fakt-HiOUTNPQ  
 NEG-borrowed REFL Eva FAKT DET.PPI book  
 Intended: 'Isn't it really the case that Eva has borrowed a book?'  
 d. ??A Eva si fakt NE-PŮJČILA nějakou knížku? ??fakt-LoOUTNPQ  
 and Eva REFL FAKT NEG-borrowed DET.PPI book  
 Intended: 'Isn't it really the case that Eva has borrowed some book?'

*Fakt* is proposed to loosen the ordering source of  $\Box_{\text{EVI}}$  with the effect of widening the quantificational domain of the modal so that it includes stereotypical as well as less stereotypical worlds. In (39), the speaker finds the current contextual evidence for  $\neg p$  hard to believe and by asking a *fakt*-INNNPQ invites the addressee to consider (or reconsider) the evidence for  $\neg p$ , even in the less stereotypical worlds before answering whether  $p$  or  $\neg p$  holds. The meaning of (39) is in (40), where *nic k jídlu* is simplified to denote FOOD.

- (39) *B offers food to A several times. A says no thank you each time. B asks:*  
 B: Tak fakt ne-chceš nic k jídlu? fakt-INNPQ  
 so FAKT NEG-want anything.NCI to eat  
 'So you really don't want anything to eat?' (Kopřivová et al. 2017)

$$(40) \quad \llbracket (39) \rrbracket =$$

At-issue:  $\{\lambda w. \text{WANT}(A, \text{FOOD})(w), \lambda w. \neg \text{WANT}(A, \text{FOOD})(w)\}$   
 Non-at-issue:  $\lambda w. [\forall w' \in \max_{o'(w)}(\text{EVI}_B(w)) [\neg \text{WANT}(A, \text{FOOD})(w') = 1]]$

We point out for completeness that the felicity of *fakt* in Czech PQs is not dependent on the presence of speaker bias. In (39), speaker A presumably does not have any prior belief about B wanting food or not. This is in contrast to some of the previous analyses of its English counterpart *really*, which stressed the connection between the particle and speaker bias (Romero & Han 2004; Goodhue 2022; Giannakidou & Mari 2025; but see Domaneschi et al.’s 2017 experimental evidence for *really*-PQs being felicitous even when the speaker has no prior belief about *p*).

Similarly to the Russian particle *razve* (‘really’), *fakt* in LoNPQs occurs with both NCIs and PPIs (Repp & Geist 2025), so it is not tied to a specific polarity. The authors suggest that *razve* is a cue used to identify the presence of VERUM or FALSUM. Analogously, *fakt* can be thought of as a cue for contextual bias.

#### 4.4 Information structure

In Section 3.3 we showed that OUTNPQs (especially LoOUTNPQs) are restricted with respect to information structure. In the analysis developed in Section 4.1.3, we assumed that in OUTNPQs, the non-at-issue modal statement is focused. Since the focus on the non-at-issue content is obligatory, it is predicted to be overtly encoded: either by fronting of the negated verb (in HiOUTNPQs) or by placing sentence stress on the negated verb (in LoOUTNPQs). These operations are not driven by the satisfaction of formal features only, but rather by the exhaustivity inference; see Horvath (2010) for a similar argument about focus fronting being driven by contrast in Hungarian. In LoOUTNPQs, the focus must strictly fall on the negated verb and not on another constituent – this is shown in (41).

- (41) A: Allegedly, Eva bought a new car.  
 B<sub>1</sub>: A [Petr]<sub>CT</sub> si náhodou [NE-KOUPIL]<sub>F</sub> nějaké auto? LoOUTNPQ  
 and Petr REFL NÁHODOU NEG-bought DET.PPI car  
 ‘And what about Petr, did he buy some car, by any chance?’  
 B<sub>2</sub>: #A [Petr]<sub>CT</sub> si náhodou ne-koupil nějakou [MOTORKU]<sub>F</sub>? LoOUTNPQ  
 and Petr REFL NÁHODOU NEG-bought DET.PPI motorbike  
 ‘And what about Petr, did he buy some motorbike, by any chance?’

Moreover, LoOUTNPQs are felicitous only when they contain a contrastive topic, so their declarative word order is licensed by the presence of a contrastive topic, not by conveying contextual bias. The contrastive topic requirement is shown in (42), where the context must contain a contrasting alternative to the PQ-initial element.

- (42) A: Allegedly, Petr didn’t buy any car.  
 B<sub>1</sub>: A [Eva]<sub>CT</sub> si náhodou [NE-KOUPILA]<sub>F</sub> nějaké auto? LoOUTNPQ  
 and Eva REFL NÁHODOU NEG-bought DET.PPI car  
 ‘And what about Eva, did she buy some car, by any chance?’  
 B<sub>2</sub>: #A Petr si náhodou [NE-KOUPIL]<sub>F</sub> nějaké auto? LoOUTNPQ  
 and Petr REFL NÁHODOU NEG-bought DET.PPI motorbike  
 ‘And did Petr buy a motorbike, by any chance?’

INNNPQs and MEDNPQs are predicted to be freer in the encoding of information structure. The negated verb can be focused (43), but other elements can be focused as well (44).

- (43) A: Slyšela jsem, že Eva si prý ne-koupila auto.  
 heard AUX that Eva REFL allegedly NEG-bought car

- ‘I heard that Eva allegedly didn’t buy a car.’
- B: Eva si fakt [NE-KOUPILA]<sub>F</sub> {žádné / nějaké} auto? INNNPQ/MEDNPQ  
 Eva REFL FAKT NEG-bought DET.NCI DET.PPI car  
 ‘Eva really didn’t buy any / some car?’
- (44) A: Slyšela jsem, že Eva si prý ne-koupila škodovku.  
 heard AUX that Eva REFL allegedly NEG-bought Skoda-car  
 ‘I heard that Eva allegedly didn’t buy a Skoda.’
- B: Eva si fakt ne-koupila {žádné / nějaké} [AUTO]<sub>F</sub>? INNNPQ/MEDNPQ  
 Eva REFL FAKT NEG-bought DET.NCI DET.PPI car  
 ‘Eva really didn’t buy any / some car?’

Let us point out for completeness that HiOUTNPQs are not as restricted as LoOUTNPQs when it comes to information structure: (i) they do not require a contrastive topic (in fact, they cannot express a contrastive topic, since they are interrogative), and (ii) they allow for focusing of other elements than the negated verb itself – since the focus on the verb in HiOUTNPQs is encoded by fronting only, another focus is free to be encoded prosodically by attracting sentence stress, see (45).

- (45) A: Allegedly, Eva bought a new car.  
 B: [Ne-koupila]<sub>F</sub> si náhodou Eva nějakou [MOTORKU]<sub>F</sub>? HiOUTNPQ  
 NEG-bought REFL NÁHODOU Eva DET.PPI motorbike  
 ‘Didn’t Eva buy some motorbike, by any chance?’

## 4.5 Embedded polar questions with negation

Examples (46), (47) and (48) show that INNNPQs, MEDNPQs and OUTNPQs, respectively, might be embedded while imposing the same licensing/compatibility conditions of polarity items and particles as their matrix counterparts.

- (46) Embedded INNNPQ:  
 Eva se zeptala, jestli Petr fakt ještě (?? náhodou) ne-koupil žádný  
 Eva REFL asked if Petr FAKT JEŠTĚ.NPI NÁHODOU NEG-bought DET.NCI  
 dárek.  
 gift  
 ‘Eva asked whether it is still the case that Petr really didn’t buy any gift.’
- (47) Embedded MEDNPQ:  
 Eva se zeptala, jestli Petr fakt (?? ještě) (?? náhodou) ne-koupil nějaký  
 Eva REFL asked if Petr FAKT JEŠTĚ.NPI NÁHODOU NEG-bought DET.PPI  
 dárek.  
 gift  
 ‘Eva asked whether Petr really didn’t buy some gift.’
- (48) Embedded OUTNPQ:  
 Eva se zeptala, jestli Petr náhodou (?? ještě) (?? fakt) ne-koupil nějaký  
 Eva REFL asked if Petr NÁHODOU JEŠTĚ.NPI FAKT NEG-bought DET.PPI  
 dárek.  
 gift  
 ‘Eva asked whether Petr bought some gift by any chance.’

Importantly, the bias in the embedded NPQs is tied to the attitude holder, not to the speaker of

the PQ. In (46) and (47), the attitude holder Eva is the one witnessing evidence for  $\neg p$ , while in (48) Eva considers both  $p$  and  $\neg p$  as accessible. The non-at-issue meanings of the embedded NPQs in (46), (47) and (48) are captured in (49), respectively. In all three cases the bias gets accommodated locally, just as non-at-issue content would under verbs like *ask* (Karttunen 1973). We take this to be evidence for the claim that bias is non-at-issue rather than at-issue.

(49) Non-at-issue content of (46):

$$\lambda w . [\forall w' \in \max_{o'(w)}(\text{EVI}_{\text{EVA}}(w))][\neg \text{BOUGHT}(\text{PETR}, \text{CAR})(w') = 1]]$$

Non-at-issue content of (47):

$$\lambda w . [\neg \forall w' \in \max_{o'(w)}(\text{EVI}_{\text{EVA}}(w))][\text{BOUGHT}(\text{PETR}, \text{CAR})(w') = 1]]$$

Non-at-issue content of (48):

$$\lambda w . [\exists w' \in \max_{o'(w)}(\text{EPI}_{\text{EVA}}(w))][\neg \text{BOUGHT}(\text{PETR}, \text{CAR})(w') = 1]] \wedge$$

$$\lambda w . [\exists w' \in \max_{o'(w)}(\text{EPI}_{\text{EVA}}(w))][\text{BOUGHT}(\text{PETR}, \text{CAR})(w') = 1]]$$

## 4.6 Predictions

Here we summarize the predictions our account of three negation reading in Czech PQs makes:

(50) Obligatory biases

- a. INNNPQs convey strong contextual bias for  $\neg p$
- b. MEDNPQs convey weak contextual bias for  $\neg p$
- c. OUTNPQs convey speaker anti-bias

(51) Polarity items

- a. NCIs licensed only in INNNPQs
- b. NPIs, including *ještě* ('still'/'yet'), licensed only in INNNPQs
- c. PPIs in scope of negation allowed in only MEDNPQs and OUTNPQs

(52) Particles

- a. *náhodou* ('by any chance') compatible only with OUTNPQs
- b. *fakt* ('really') compatible only with INNNPQs and MEDNPQs

(53) Information structure

- a. obligatory focus on the negated verb only in OUTNPQs
- b. obligatory contrastive topic only in LoOUTNPQs

In the following section we present novel experimental data testing these predictions. The goal is to provide experimental evidence for the distinction of the three negation readings. First, we describe the general nature of the experiment, its procedure and participants. Then, we present two experiments with their designs, predictions and results separately. The results are discussed together for the two experiments in relation to the predictions of the proposed analysis in Section 6.

## 5 Experimental evidence

We ran a naturalness judgment task where participants rated how natural a PQ sounds in a short context.<sup>14</sup> The experimental set-up consisted of 12 experiments; each of them had its own design, but the task and rating scale remained the same. There were 86 items in

<sup>14</sup>The preregistration, experimental materials, results and analyses can be obtained through this view only link: [https://osf.io/at7sh/overview?view\\_only=512b3557986e4204a18a93d91ab1f4da](https://osf.io/at7sh/overview?view_only=512b3557986e4204a18a93d91ab1f4da).

total. Stimuli from the individual experiments served as fillers for one another, which ensured variability of the stimuli. They were distributed on lists as per the Latin square design and the order within each list was pseudorandomized. The task was run online on the platform L-Rex (Starschenko & Wierzba 2026).

The general structure of the stimuli was the same throughout all the individual experiments. The structure is exemplified in (54) with a PPQ. Each stimulus had the form of a short conversation between two people, A and B, where A provided context and B reacted to it by asking a question. The context consisted of two clauses, where the second one always contained some hedging (for example the particle *prý* ‘allegedly’ or the parenthetical *myslím* ‘I think’ or *zdá se* ‘it seems’). The question generally consisted of only one clause. About 30% of the items involved PPQs, and about 70% included NPQs. The stimuli were pre-recorded and presented as audio.

- (54) A: Viktor vyjel na filmový festival a *prý* tam viděl Brada Pitta. context  
 Viktor went on film festival and allegedly there saw Brad Pitt  
 ‘Viktor went to a film festival and allegedly he saw Brad Pitt there.’  
 B: On fakt Brada Pitta VIDĚL? question  
 he really Brad Pitt saw  
 ‘He really saw Brad Pitt?’

A link to the experimental environment was distributed among participants as well as a consent form and instructions for the task. There were two test stimuli where the participants had a chance to try out the rating. The scale had 7 points, where 1 = completely unnatural and 7 = completely natural. The task took them approximately 30 minutes. In total, 71 participants took part in the experiment. They were all Czech native speakers. As a reward for their participation, they received course credits.

The analysis of the results was done in RStudio (Posit team 2025). We fitted cumulative link mixed effects models (CLMMs) using the package *ordinal* and function *cglm*, which are suitable for ordinal data (Christensen 2022). We ran post-hoc tests using the package *emmeans* (Lenth 2025).

In the following subsections, we describe in more detail two experiments which were part of the larger set-up. For each experiment, we explain the design and the predictions stemming from the theoretical assumptions made earlier. We provide results for each of the experiments in the relevant subsection. Experiment 1, described in Section 5.1, teases apart medial and inner negation by manipulating the negation reading in the presence of the NPI *ještě* (‘still’/‘yet’). Experiment 2, described in Section 5.2, teases apart OUTNPQs from MEDNPQs and INNNPQs by manipulating information structure and the locus of sentence stress.

## 5.1 Experiment 1 – NPI licensing

### 5.1.1 Design & predictions

Experiment 1 aimed to distinguish between MEDNPQs from INNNPQs by testing their ability to license NPIs, operationalized through the presence of the NPI *ještě* (‘still’/‘yet’). The manipulated variable was NEGATION READING with two levels: medial vs. inner, and it was proxied by polarity items in the PQ: medial negation by the PPI *nějaký* (‘some.DET.PPI’), while inner negation by the NCI *žádný* (‘no.DET.NCI’). NEGATION READING was manipulated within items and within subjects. Two more factors were controlled for across the items: (i) the contextual cue uttered by A always suggested  $\neg p$  (that is, the evidence for  $\neg p$  weak weak due to the hedg-

ing), which was expected to be necessary for both the negation readings to be felicitous,<sup>15</sup> and (ii) all the predicates in the PQs were telic. An example item is in (55); altogether there were 8 items in this experiment. Note that in this experiment, the negation occurring alongside a PPI, as in (55B<sub>2</sub>), had to be interpreted as medial, not as outer; in the latter case, a contrastive topic would have been required, as predicted in (53b), which was not the case here.

- (55) A: Petr je sice v kuchyni, ale vypadá to, že kouká jen do mobilu.  
 Petr is though in kitchen but seems it that looks only into phone  
 ‘Even though Petr is in the kitchen, it seems that he is only on his phone.’  
 B<sub>1</sub>: On nám ještě ne-připravil žádnou SVAČINU? INNNPQ  
 he us JEŠTĚ.NPI NEG-prepared DET.NCI snack  
 ‘Is it still the case that he hasn’t prepared us any snack?’  
 B<sub>2</sub>: On nám ještě ne-připravil nějakou SVAČINU? MEDNPQ  
 he us JEŠTĚ.NPI NEG-prepared DET.PPI snack  
 ‘Is it still the case that he hasn’t prepared us some snack?’

According to the analysis proposed earlier in the paper, inner negation operates on the at-issue level and its domain is the clause. Medial negation operates on the non-at-issue level and so leaves the at-issue content intact. The analysis predicts that only inner negation can license the NPI *ještě* as per (51b). If the analysis is on right track, it is expected that INNNPQs with *ještě* are rated as more natural than MEDNPQs with *ještě*.

### 5.1.2 Results

The raw results of experiment 1 are shown in Figure 1a. We ran a CLMM with the fixed factor NEGATION READING and random intercepts and slopes for participants and items. We used treatment coding with inner negation as the baseline. There was a main effect of NEGATION READING ( $p < .001$ ,  $z = -3.775$ ), which shows a clear preference of INNNPQs to host *ještě* (‘still’/‘yet’) in comparison to MEDNPQs. The emmeans results are shown in Figure 1b.

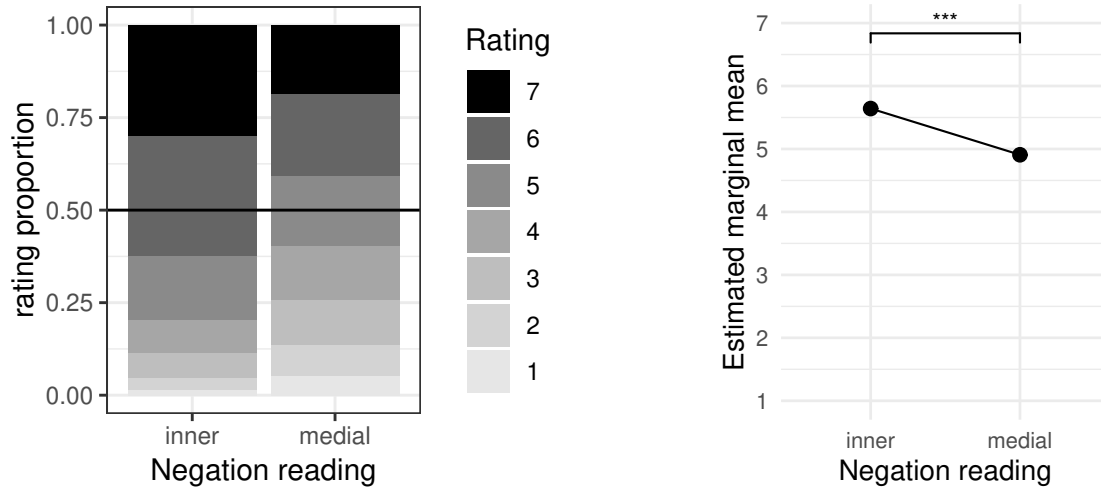
## 5.2 Experiment 2 – focus placement

### 5.2.1 Design & predictions

Experiment 2 was designed to further distinguish between (low) OUTNPQs on the one hand and MEDNPQs/INNNPQs on the other. The two manipulated variables were NEGATION READING with three levels: outer vs. medial vs. inner; and FOCUSED CONSTITUENT with two levels: negated verb vs. object; resulting in a  $3 \times 2$  design. NEGATION READING was manipulated between items and within subjects, while FOCUSED CONSTITUENT was manipulated within items and within subjects.

Focused constituents in Czech tend to carry sentence stress: a word (negated verb or the last element of an object NP) carries sentence stress when it is placed in the utterance-final position, which in turn attracts sentence stress by default in Czech and is common for marking focus (Daneš 1957). The information structure of the PQ, including which constituent was focused, was manipulated contextually. NEGATION READING was proxied by a combination of a PPI/NCI and a certain particle. Outer negation was cued by the PPI *nějaký* (‘some.DET.PPI’) and

<sup>15</sup>We assumed different categories of evidence (Willett 1988). Directly witnessed evidence (visual/auditory) was assumed to be compatible only with strong contextual bias (thus, INNNPQs only), while indirectly witnessed evidence (reported/inferred) was assumed to be compatible with strong and weak contextual bias (thus, both MEDNPQs and INNNPQs).



(a) Raw results of exp. 1. (the horizontal line cuts through the median)

(b) Emmeans results of exp. 1

Figure 1: Results of experiment 1: naturalness of INNNPQs vs. MEDNPQs with the NPI *ještě* ('still'/'yet')

the particle *náhodou* ('by any chance').<sup>16</sup> There were 8 items with outer negation; exemplified in (56). In this experiment, all OUTNPQs involved low negation and a contrastive topic.<sup>17</sup>

(56) Focused negated verb:

A<sub>1</sub>: Sára se spolužačkou šly psát test z biologie, ale její spolužačka prý Sára with classmate went write test from biology but her classmate allegedly ne-měla tužku.  
NEG-had pen

'Sára and her classmate were about to write a biology test, but allegedly the classmate didn't have any pen.'

B<sub>1</sub>: A [Sára]<sub>CT</sub> náhodou nějakou tužku [NE-MĚLA]<sub>F</sub>? OUTNPQ  
and Sára NÁHODOU DET.PPI pen NEG-had  
'And what about Sára, did she have some pen, by any chance?'

Focused object:

A<sub>2</sub>: Sára se spolužačkou šly psát test z biologie, ale její spolužačka prý Sára with classmate went write test from biology but her classmate allegedly ne-měla nic na psaní.  
NEG-had nothing.NCI for writing

'Sára and her classmate were about to write a biology test, but allegedly the classmate didn't have anything to write with.'

B<sub>2</sub>: A [Sára]<sub>CT</sub> náhodou ne-měla aspoň nějakou [TUŽKU]<sub>F</sub>? OUTNPQ  
and Sára NÁHODOU NEG-had at-least DET.PPI pen  
'And what about Sára, did she have some pen, by any chance?'

<sup>16</sup>We believe that *náhodou* could not have been received its canonical, non-particle reading meaning 'accidentally' in the contexts of the experimental stimuli.

<sup>17</sup>In an auxiliary experiment, we tested LoOUTNPQs with vs. without a contrastive topic. LoOUTNPQs with a contrastive topic were significantly more natural than those without it ( $p < .001$ ,  $z = 9.894$ ; median rating 5 vs. median rating 1, respectively).



Medial negation was cued by the PPI *nějaký* ('some.DET.PPI') and the particle *fakt* ('really'). There were 8 items with medial negation; exemplified in (57).

(57) Focused negated verb:

- A<sub>1</sub>: Jitka jela s Járrou k moři a Jára si prý zapomněl zabalit opalovací  
Jitka went with Jára to sea and Jára REFL allegedly forgot pack sunscreen  
krém.  
cream  
'Jitka and Jára went to the seaside and Jára allegedly forgot to pack sunscreen.'  
B<sub>1</sub>: On si fakt nějaký krém [NE-ZABALIL]<sub>F</sub>? MEDNPQ  
he REFL FAKT DET.PPI sunscreen NEG-packed  
'He really didn't pack some sunscreen?'

Focused object:

- A<sub>2</sub>: Jitka jela s Járrou k moři a Jára si prý zapomněl zabalit hygienickou  
Jitka went with Jára to sea and Jára REFL allegedly forgot pack hygienic  
taštičku.  
bag  
'Jitka and Jára went to the seaside and Jára allegedly forgot to pack a toiletry bag.'  
B<sub>2</sub>: On si fakt ne-zabalil aspoň nějaký [KRÉM]<sub>F</sub>? MEDNPQ  
he REFL FAKT NEG-packed at-least DET.PPI sunscreen  
'He really didn't pack at least some sunscreen?'

Inner negation was cued by the NCI *žádný* ('no.DET.NCI') and the NPI *ještě* ('still'/'yet'). There were 8 items with inner negation; exemplified in (58).

(58) Focused negated verb:

- A<sub>1</sub>: Pan Novák chtěl oblečení do divadla, ale prodavačka mu snad zatím  
Mr Novák wanted clothes to theatre but shop-assistant him probably yet  
ne-nabídla ani košili.  
NEG-offered not-even shirt  
'Mr. Novák wanted clothes to wear to the theatre, but the shop assistant probably  
hasn't even offered him a shirt.'  
B<sub>1</sub>: Ona mu ještě žádnou košili [NE-NABÍDLA]<sub>F</sub>? INNNPQ  
she him JEŠTĚ.NPI DET.NCI shirt NEG-offered  
'Is it still the case that she hasn't offered him any shirt?'

Focused object:

- A<sub>2</sub>: Pan Novák chtěl oblečení do divadla, ale prodavačka mu snad zatím  
Mr Novák wanted clothes to theatre but shop-assistant him probably yet  
nic ne-nabídla.  
nothing.NCI NEG-offered  
'Mr. Novák wanted clothes to wear to the theatre, but the shop assistant probably  
hasn't offered him anything.'  
B<sub>2</sub>: Ona mu ještě ne-nabídla žádnou [KOŠILI]<sub>F</sub>? INNNPQ  
she him JEŠTĚ.NPI NEG-offered DET.NCI shirt  
'Is it still the case that she hasn't offered him any shirt?'

OUTNPQs occurred in contexts which did not suggest  $p$  or  $\neg p$ , while both MEDNPQs and INNNPQs occurred in contexts suggesting  $\neg p$ . The MEDNPQs and INNNPQs in this experiment contained all-given material (in the sense of e.g., Rochemont 2016; i.e., every element of the

PQ had an anaphoric predecessor in the context).

The proposed analysis predicts that in OUTNPQs, the negated verb must be focused, as per (53a). The focus is obligatory, triggers the EXH operator, which, in turn, outputs the anti-bias observed in OUTNPQs. In LoOUTNPQs the focus is encoded by sentence stress on the negated verb. There is no such requirement in MEDNPQs or INNNPQs. If the analysis is on the right track, it is expected that LoOUTNPQs with a focused negated verb are rated as more natural than LoOUTNPQs with a focused object. As for MEDNPQs and INNNPQs, it is expected that their information structure is free and so there would not be a rating difference between the conditions with a focused negated verb as opposed to those with a focused object.

### 5.2.2 Results

The raw results of experiment 2 are shown in Figure 2. We ran a CLMM with two fixed factors NEGATION READING and FOCUSED CONSTITUENT and random intercepts and slopes for participants and items. We used helmert coding for NEGATION READING and sum coding for FOCUSED CONSTITUENT. There was a significant interaction of NEGATION READING and FOCUSED CONSTITUENT ( $p < .001$ ,  $z = 6.01$ ), which shows that OUTNPQs follow a different trend from MEDNPQs/INNNPQs. The emmeans results are in Figure 3.

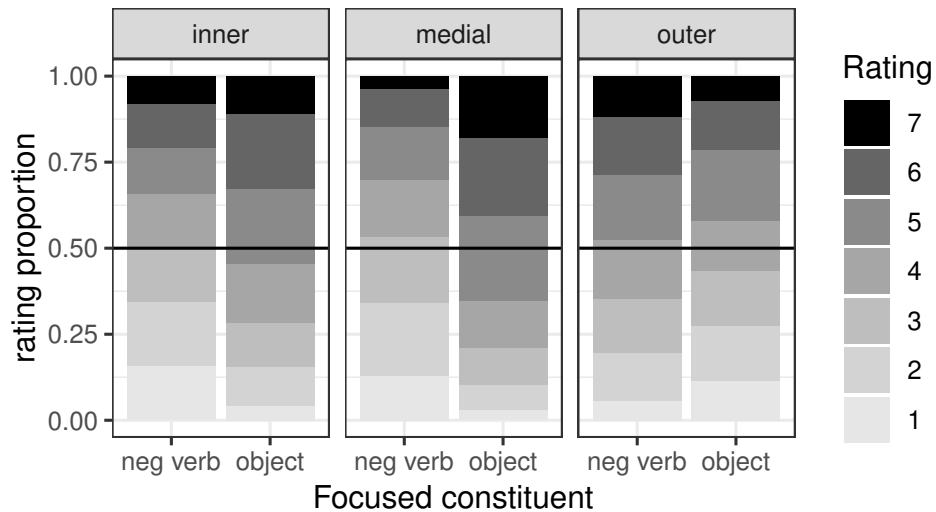


Figure 2: Raw results of experiment 2: naturalness of INNPNQs, MEDNPQs and OUTNPQs with a focused negated verb vs. a focused object (the horizontal line cuts through the median ratings)

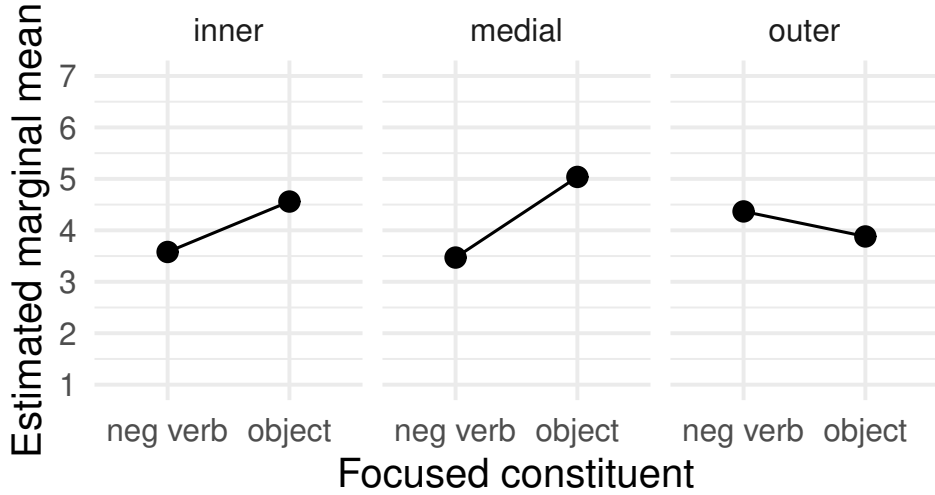


Figure 3: Emmeans results of experiment 2: naturalness of INNNPQs, MEDNPQs and OUTNPQs with a focused negated verb vs. a focused object

## 6 Discussion

According to the analysis we put forth in Section 4, there are three negation readings that can be discerned in Czech PQs. While maintaining the general function of logical negation, we proposed that the negative operator can operate on different layers of meaning and can interact with different question bias modals. The so-called inner negation is operative on the at-issue layer and scopes under the contextual bias modal. Medial negation is operative on the non-at-issue layer and scopes above the contextual bias modal and modifies it. Outer negation is likewise operative on the non-at-issue layer, triggers and modifies the speaker bias modal. The resulting modal statement is strengthened by exhaustification.

Such an analysis predicts that only inner negation can license elements like standard sentential negation, whereas medial and outer negation cannot, as they reside on a different layer. The elements predicted to be licensed by inner negation but not the other two readings are NCIs and NPIs like *ještě* ('still'/'yet'), as per (51a) and (51b). Section 6.1 discusses the results of experiment 1, where inner negation was distinguished from medial based on its ability to license the NPI *ještě*.

Further, the analysis predicts that OUTNPQs require focus on the negated verb, as per (53a). This requirement does not hold for MEDNPQs or INNNPQs. Section 6.2 discusses the results of experiment 2, where OUTNPQs were distinguished from MEDNPQs and INNNPQs based on their requirement to focus the verb.

### 6.1 Medial vs. inner negation

The results of experiment 1 support the claim that inner and medial negation impose different licensing conditions. In particular, *ještě*-INNNPQs were rated as significantly more natural than *ještě*-MEDNPQs. We interpret this result as a clear difference between the two readings.

Even though the main effect of NEGATION READING was significant, one can still wonder about the relatively high absolute rating of *ještě*-MEDNPQs, which is discernible in Figure 1a. We offer two possible explanations of why some *ještě*-MEDNPQs got high ratings. Some of the participants might have interpreted the negative marker as inner negation and rescued

the sentence by ignoring the surface scope of the negated verb and the PPI ( $\neg > \text{PPI}$ ) in favor of the inverse scope ( $\text{PPI} > \neg$ ). Recall that like sentential negation, inner negation can be outscoped by PPIs. Yet, the interpretation of a PPI outscoping the negation in these contexts seems rather implausible.

Alternatively, some participants might have judged these structures as natural under the impression of a grammatical illusion, that is, they fleetingly considered the ungrammatical *ještě*-MEDNPQ to be natural. This phenomenon has been recently tested by Lacina & Dočekal (2026) in Czech, where sentential negation in an embedded clause was shown to serve as an illusory licenser to NPIs and NCIs in the matrix clause. In the present case, the participants might have in fact entertained the expected medial interpretation (due to negation scoping over the PPI) and interpreted it as an illusive licenser for *ještě*.

Future research might support the distinction between INNNPQs vs. MEDNPQs by testing their naturalness in different contexts. According to the predictions in (50a) and (50b), MEDNPQs are expected to sound natural only in contexts with weak evidence for  $\neg p$ , while INNNPQs are expected to sound natural in contexts with weak or strong evidence for  $\neg p$ . Another possibility would be to test these PQs with other NPIs in them, such as *vůbec* ('at all') or *ani* ('not even'). Under the present analysis, INNNPQs are expected to license *vůbec* and *ani*, while MEDNPQs not.

## 6.2 Outer vs. medial/inner negation

Having discussed the distinction between medial and inner negation, let's now focus on how to distinguish outer negation from them. LOOUTNPQs were predicted to exhibit restricted information structure in that focus would obligatorily fall on the negated verb and not on another constituent. MEDNPQs and INNNPQs were predicted to be freer in this sense, where focus could fall essentially on any constituent in the PQ. In experiment 3, we compared the three types of NPQs (LOOUTNPQs, MEDNPQs and INNNPQs) while manipulating which constituent was focused.

The results, namely the significant interaction of NEGATION READING and FOCUSED CONSTITUENT, showed that OUTNPQs follow a different pattern from MEDNPQs and INNNPQs. Czech native speakers prefer focus on the negated verb more in LOOUTNPQs than in INNNPQs or MEDNPQs, and they reject focus on the object more in LOOUTNPQs than in INNNPQs or MEDNPQs. Such results provide support for the present analysis.

When zooming in on the subset of LOOUTNPQs only, we see a numerical preference for the focused negated verb, which speaks in favor of the account developed here. One might ask about the relatively high rating of the focused object condition LOOUTNPQs (which was rather unexpected). We offer two possible explanations. The rating might have been influenced by the participants having missed or ignored the particle *náhodou* ('by any chance') in what was intended to represent LOOUTNPQs. Remember that the reading of outer negation was proxied by a PPI as well as the particle *náhodou*. However, this does not seem as a plausible explanation – in an auxiliary experiment, where *náhodou* was inserted in INNNPQs and MEDNPQs, participants consistently rated those as unnatural (their median rating was 2). We can therefore conclude that participants were sensitive to *náhodou*.

The second and more probable explanation is that in general, some participants favored the default word order (i.e., SVO) over the non-default one (i.e., SOV, where the object is scrambled). The tendency towards the default word order is evident in the results for INNNPQs and MEDNPQs – the conditions with a focused object generally received higher ratings than the conditions with a focused negated verb. Moreover, recall that both MEDNPQs and INNNPQs

contained all-given material. In all-given utterances, it seems unnatural to perform object scrambling, unless there is some additional requirement to fulfill. We hypothesize that the tendency towards the default word order is the reason why LoOutNPQs with a focused object received the relatively high rating.

## 7 Conclusion

In this paper, we focused on the semantics of Czech polar questions with negation. The verbal prefix *ne-* can be syntactically situated either high (in interrogative PQs) or low (in declarative PQs). Low negation has been previously claimed to be semantically/pragmatically ambiguous. Here, we distinguished three readings of (low) negation which we called inner, medial and outer. Each negation reading was represented by  $\neg$  operating on a different meaning tier and interacting with a different bias modal. Inner negation ( $\neg_{inn}$ ), just like standard sentential negation, operates on the at-issue meaning tier. Inner negation polar questions convey strong contextual bias for  $\neg p$ , since  $\neg_{inn}$  interacts with the contextual bias modal. Medial negation ( $\neg_{med}$ ) targets the non-at-issue content of the contextual bias modal. As a result, medial negation polar questions carry weak contextual bias for  $\neg p$ . Finally, outer negation ( $\neg_{out}$ ) triggers the speaker bias modal and, similarly to medial negation, operates on the non-at-issue tier. The modal statement is focused, which triggers the exhaustivity inference. As a result, outer negation polar questions encode speaker anti-bias. The proposed analysis treats negation and question bias in a uniform manner and allows for capturing cross-linguistic variability of question meaning. The proposed analysis was supported by results from a naturalness judgment task where participants rated how natural certain polar questions sounded. The results suggest that Czech native speakers are sensitive to the three negation readings.

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